

15.439 Assignment 1

This homework is due at 4:00 pm on February 25th at the start of class.

Please read and familiarize yourself with the expectations for how homework is to be done (e.g. the size of the group, contribution required of all group members, allowable methods, etc).

Data Scientists and Analysts are often tasked to clean and analyze datasets.

Imagine you are asked by your Portfolio Manager at the hedge fund you currently work at to evaluate a new third party firm, WKKWNT, LLC (aka **We KnotKnow NeThing**, LLC.). WKKWNT specializes in the application forecasting prices of financial instruments. This firm has developed a proprietary system, called “4sight”, to forecast prices of certain instruments.

To demonstrate the effectiveness of their forecasting system, WKKWNT sent the attached sample dataset. The dataset includes signal values generated by the 4sight system as well as historical prices for a well-known broad market ETF.

Your portfolio manager has asked you to:

1. Review the quality of the data, list any potential errors, and propose corrected values. Please list each quality check error and each correction applied to the data in its own spreadsheet. Do not correct the errors in the data spreadsheet but in a spreadsheet where we can quickly and easily see (on a standalone basis) the errors you fixed and how you found them. Credit will be given for the errors you caught and fixed; points will be deducted for errors that you “fixed” that weren’t actually errors. Points will be a function of the total errors that you caught.
2. Please analyze the signal’s effectiveness or lack thereof in forecasting the ETF price, using whatever metrics you think are most relevant. Please explain why you chose the metrics that you did. This is NOT meant to be a dissertation or long and involved. What can you say and see relatively quickly. This will serve as a useful baseline for yourself as we start the course.
3. (Extra credit) Write a 1-2 paragraph summary for the portfolio manager addressing your observations about the efficacy and believability of the product, and recommendation for next steps.

My expectation is that this homework should take you approximately 2-3 hours but that is dependent, ultimately, on your knowledge of financial data and your ability.